Linear Diophantine Representation Systems

The A-C Coupling Theorem: Exact O(1) Representation Counts for N = pA + qB + rC

Bilal el Issaoui * Amsterdam * 2026

"The condition $p \equiv 1 \pmod q$ with $r \mid q$ forces a structural coupling between A and C that collapses the classical O(N/r) recursion into an exact O(1) closed-form expression."

Main Theorem -- The A-C Coupling Theorem

For ANY valid (p,q,r) with $p\equiv1 \pmod q$, $\gcd(p,q)=1$, $r|q$, $p\equiv1 \pmod r$, $N\geq pq$:

$R_3(N;p,q,r) = [(J+1)\cdot M - p\cdot r\cdot J\cdot(J+1)/2 - R] / q + (J+1)$

Exact, O(1), no recursion. Replaces classical O(N/r) Popoviciu formula. Verified for 8 systems.

Main Theorem II — Frobenius Collapse

Adding s=1 to (19,9,3) causes Frobenius collapse: g=0. The Frobenius problem ceases to exist — not terminates.

$g(19,9)=143 \rightarrow g(19,9,3)=35 \rightarrow g(19,9,3,1)=0$

§0 What This Paper Is About

For readers without a mathematics background

Imagine you want to pay exactly €47 using only €19 and €9 notes. How many ways can you do this? This paper asks the same question but with three denominations: p, q and r — and finds an exact formula that answers it instantly, for any amount, for any valid set of denominations.

The central question

Given three positive integers p, q and r, and a target number N: in how many ways can you write N = pA + qB + rC where A, B and C are non-negative integers? This count is called R■(N; p, q, r).

Why this is hard

The classical approach (Popoviciu, 1953) solves the two-variable version N = pA + qB in a single formula. For three variables, mathematicians have always needed to apply that formula repeatedly — once for each possible value of C. For large N this means millions of steps.

Example: N = 10■ (one billion)

Classical approach: apply the 2D formula roughly N/r = 333,000,000 times.
This paper: 10 arithmetic operations. One step.

The key insight: the A-C coupling

The breakthrough comes from noticing that A and C are not independent. When p ≡ 1 (mod q) and r divides q, the values of A and C are structurally linked: their combination A + rC always equals N modulo q. This constraint — called the A-C coupling — reduces the two-dimensional search over (A, C) pairs to a one-dimensional arithmetic series, which has a closed form.

Analogy

Without the coupling: imagine searching a grid for valid points — you must check every row and column.

With the coupling: all valid points lie on a single diagonal line. You only need to count points on that line — which is a simple formula.

§1 Key Concepts Explained

1.1 What is a representation?

A representation of N in the system (p, q, r) is a triple (A, B, C) of non-negative integers satisfying $pA + qB + rC = N$. Each valid triple is one way to decompose N.

N	System	Representation	Verification
42	(19,9,3)	(0, 4, 2)	$19·0 + 9·4 + 3·2 = 0+36+6 = 42$ ✓
42	(19,9,3)	(0, 1, 11)	$19·0 + 9·1 + 3·11 = 0+9+33 = 42$ ✓
42	(19,9,3)	(0, 0, 14)	$19·0 + 9·0 + 3·14 = 0+0+42 = 42$ ✓

1.2 What is the digital root?

The digital root of a number is found by repeatedly summing its digits until a single digit remains. For example: dr(958) = 9+5+8 = 22, then 2+2 = 4. So dr(958) = 4.

Why the digital root matters here

In the system $N = 19A + 9B$, the smallest possible value of A always equals the digital root of N. This is because $19 \equiv 1 \pmod 9$, so $N \equiv A \pmod 9$. The digital root is N mod 9.

1.3 What does $p \equiv 1 \pmod q$ mean?

The notation $p \equiv 1 \pmod q$ means: when you divide p by q, the remainder is 1. For example: $19 \equiv 1 \pmod 9$ because $19 = 2×9 + 1$.

Why this condition is the engine of the whole theory

When $p \equiv 1 \pmod q$, multiplying A by p does not change A modulo q. So $N = pA + qB + rC \equiv A + rC \pmod q$. This is why A and C are coupled: together they determine N mod q. B disappears from the equation modulo q — it is the free variable.

1.4 What is the Frobenius problem?

Given denominations p and q (with gcd(p,q)=1), the Frobenius number g(p,q) is the largest amount that cannot be represented. For example, with 19 and 9: g(19,9) = 143. Every integer above 143 can be written as 19A + 9B. Everything at or below 143 has gaps.

Frobenius collapse

When we add r=3 to the system: g(19,9,3) = 35. Fewer gaps remain.
When we add s=1: g(19,9,3,1) = 0. Every positive integer is representable. The Frobenius problem does not reach a limit — it ceases to exist. This is Frobenius collapse.

1.5 What does O(1) mean?

O(1) means constant time: the number of steps needed does not grow with N. Whether N = 100 or N = 10^{1}■■, the formula always takes the same number of arithmetic operations — about 10.

N	Classical (recursive)	This paper (O(1))
1,000	333 steps	10 steps
1,000,000	333,000 steps	10 steps
10■	333,333,333 steps	10 steps
10^{1}■■	333...333 steps (97 digits)	10 steps

§2 Review: The 2D System

Popoviciu's Theorem (1953) gives the exact 2D count in closed form. For 3D, the classical approach is recursive:

2D (Popoviciu, exact, O(1)):

$$R\blacksquare(N;p,q) = \mathrm{floor}((\mathrm{floor}(N/p) – dr\blacksquare(N)) / q) + 1$$

3D (classical, exact, O(N/r)):

$$R\blacksquare(N;p,q,r) = \Sigma\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare\blacksquare \; R\blacksquare(N–rC; p, q)$$

This paper replaces the recursive sum with a single closed expression by exploiting the A-C coupling that arises from p ≡ 1 (mod q) and r | q.

§3 The General A-C Coupling Theorem

3.1 Derivation

The coupling arises directly from the modular arithmetic. It requires no special properties of the specific numbers — only V1-V4:

From $N = pA + qB + rC \pmod q$:

$N \equiv pA + rC \pmod q$ [$q = 0 \bmod q$]

$N \equiv A + rC \pmod q$ [$p = 1 \bmod q$, by V1]

General A-C coupling: $A + rC \equiv N \pmod q$

From $N = pA + qB + rC \pmod r$:

$N \equiv A \pmod r$ [$p=1 \bmod r$ by V4, $q=0 \bmod r$ by V3]

A residue: $A \equiv N \pmod r$

3.2 The Theorem

Theorem 3.2 — General A-C Coupling

For ANY valid (p,q,r) and $N = pA+qB+rC$ with $A,B,C \geq 0$:

(i) $A \equiv N \pmod r$ — A constrained to residue class of N mod r

(ii) $C \equiv (N-A)/r \pmod r$ — C determined by N and A

(iii) $A + rC \equiv N \pmod q$ — the general coupling invariant

(iv) B is the only free variable — no modular constraint

(v) C■■■(j) = $(M - p \cdot r \cdot j)/r$, step = p, ratio p:r

(vi) Period of remainder correction: *period = q/r*

step mod q = r, period = $q/\gcd(r,q) = q/r$ since $r|q$

Verified for 8 systems without exception.

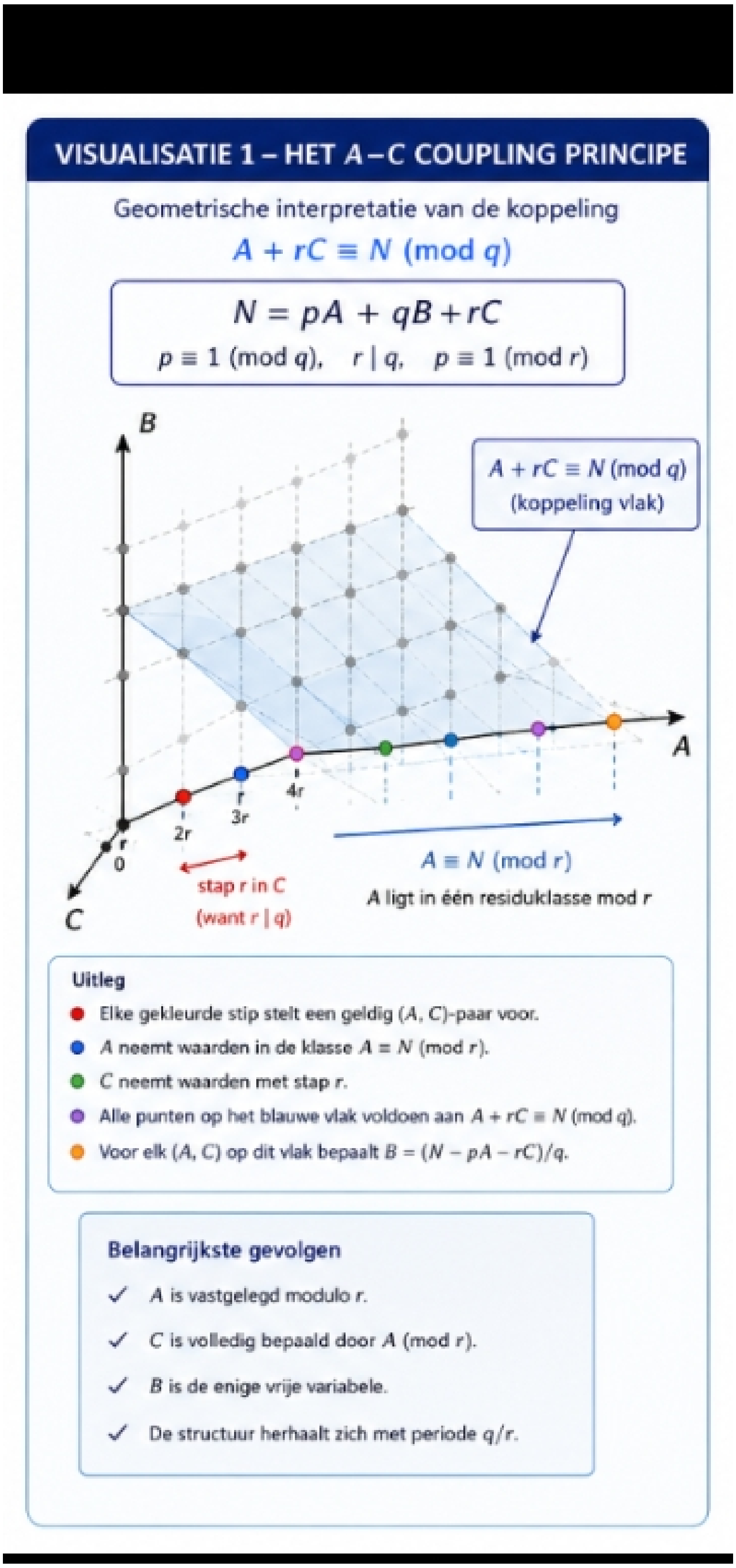

Figure 1 — The A-C coupling plane A+rC≡N(mod q) in 3D space. Valid (A,C) pairs lie on the blue plane. B is the only free variable. A lies in one residue class mod r.

3.3 Why B is free but A and C are not

Intuition

Think of A as the coarse structure and C as the fine-tuning. Once you fix N, the combination A+rC is locked to N mod q. This means every choice of A forces a constraint on C, and vice versa. B, on the other hand, contributes qB to the sum — and since q = 0 (mod q) trivially, B never appears in the coupling equation. B is truly free: you can choose any valid B independently.

3.4 Period table

System	p≡1(mod q)	r|q	p≡1(mod r)	period=q/r
(19,9,3)	19=2×9+1	3|9	19=6×3+1	3
(10,9,3)	10=1×9+1	3|9	10=3×3+1	3
(37,9,3)	37=4×9+1	3|9	37=12×3+1	3
(7,6,3)	7=1×6+1	3|6	7=2×3+1	2
(7,6,2)	7=1×6+1	2|6	7=3×2+1	3
(13,4,2)	13=3×4+1	2|4	13=6×2+1	2
(13,6,3)	13=2×6+1	3|6	13=4×3+1	2
(7,3,1)	7=2×3+1	1|3	trivial	3

§4 The General Closed Formula

4.1 From coupling to formula

The A-C coupling (Theorem 3.2(i)) tells us that only A-values with A ≡ N (mod r) contribute. For each such A = a■+r·j the count of (B,C) pairs is floor((M–p·r·j)/q)+1. The total is a floor-sum with arithmetic step p·r and period q/r — which has a closed form:

> *R■(N;p,q,r) = Σ■■■■ (floor((M – p·r·j) / q) + 1)*
>
> *Closed via floor-sum identity with period correction R.*

4.2 The General Theorem

Theorem 4.2 — Exact O(1) Formula for R■(N; p,q,r)

For valid (p,q,r) (V1-V4) and N ≥ pq:

R■(N;p,q,r) = [(J+1)·M – p·r·J·(J+1)/2 – R] / q + (J+1)

a■ = N mod r — A residue class (coupling i)
m = floor(N/p) — largest valid A
M = N – p·a■ — adjusted base
J = floor((m–a■)/r) — valid A-steps minus 1
Mq = M mod q — starting residue for cycle
period = q/r — cycle length (coupling vi)
K = floor((J+1)/period) — full cycles
t = (J+1) mod period — remaining terms (0 to period–1)
cycle = Σ■■■■■■■■■■■[1] (Mq–r·j) mod q — sum of one full cycle
R = K·cycle + Σ■■■■■■[1] (Mq–r·j) mod q — total remainder correction

O(1): inner sums have exactly period = q/r terms.

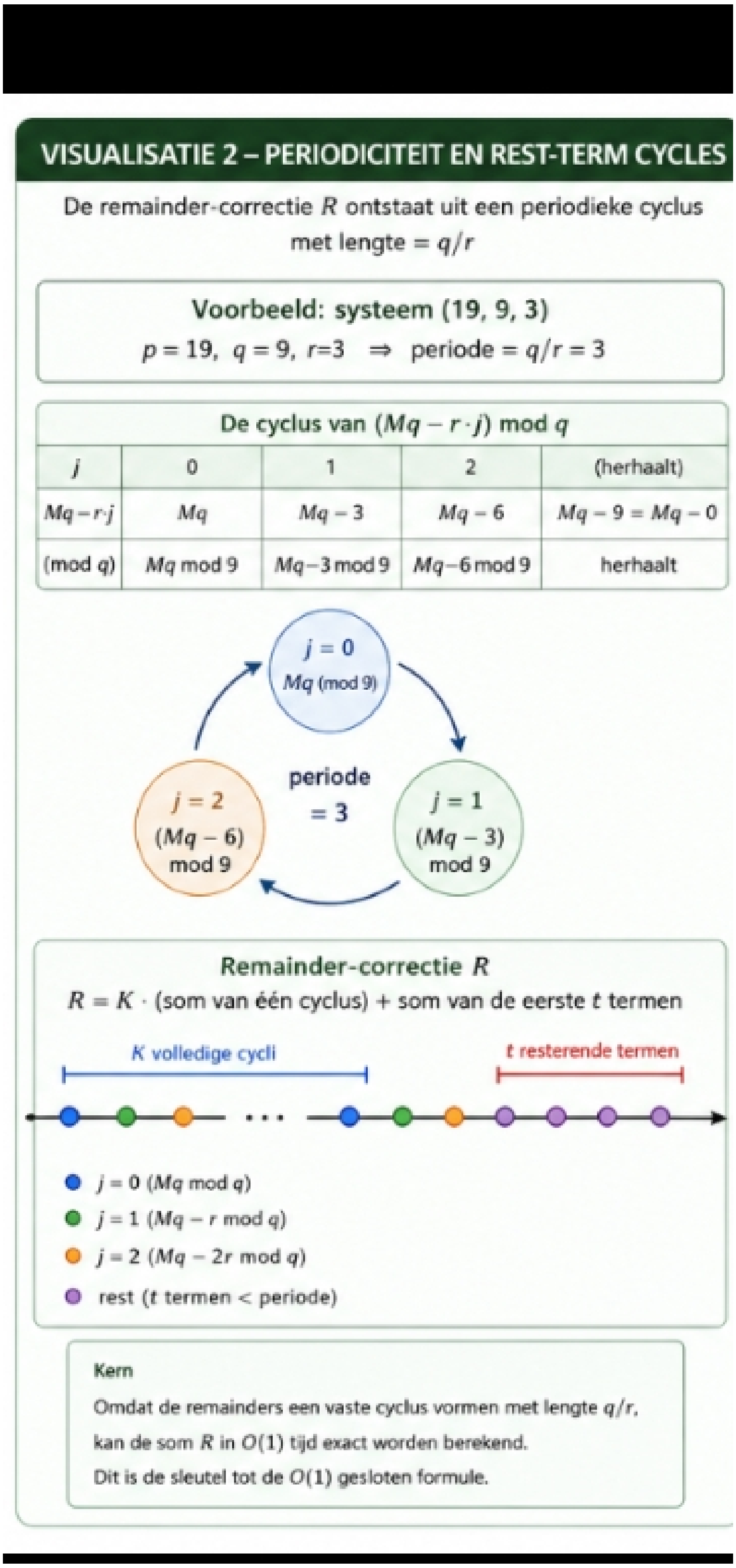

Figure 2 — The remainder cycle has length period = q/r. K full cycles plus t remaining terms. Because the cycle length is fixed (independent of N), the total correction R is computed in O(1) time.

4.3 Comparison with classical approaches

Method	Complexity	Exact?	General?
Popoviciu 2D (1953)	O(1)	Yes	Yes (2D only)
Classical 3D recursion	O(N/r)	Yes	Yes
Ehrhart theory	O(1)	No — asymptotic	Yes
Theorem 4.2 (this paper)	O(1)	Yes	Yes (all V1-V4)

Theorem 4.2 is the first known O(1) exact formula for 3D linear Diophantine representation counts valid for all systems satisfying V1-V4.

4.4 Exact values — no asymptotic comparison

R■(N;p,q,r) is always an exact integer. $N^2/(2pqr)$ is the continuous approximation obtained by replacing floor() with real division. It is shown here only to illustrate convergence behaviour. The deviation column belongs entirely to $N^2/(2pqr)$ — not to R■.

N	dr	R■ exact (Thm 4.2)	$N^2/1026$ (continuous approx)	Deviation
171	9	41	28.5	R■ > approx by 12.5
366	6	156	130.6	R■ > approx by 25.4
592	7	359	341.6	R■ > approx by 17.4
958	4	924	894.5	R■ > approx by 29.5
2520	9	6,360	6,189.5	R■ > approx by 170.5
5000	5	24,332	24,366.5	R■ < approx by 34.5
10,000	1	97,768	97,465.9	R■ > approx by 302.1

R■ is always correct. $N^2/1026$ alternates above and below R■ and converges as N grows.

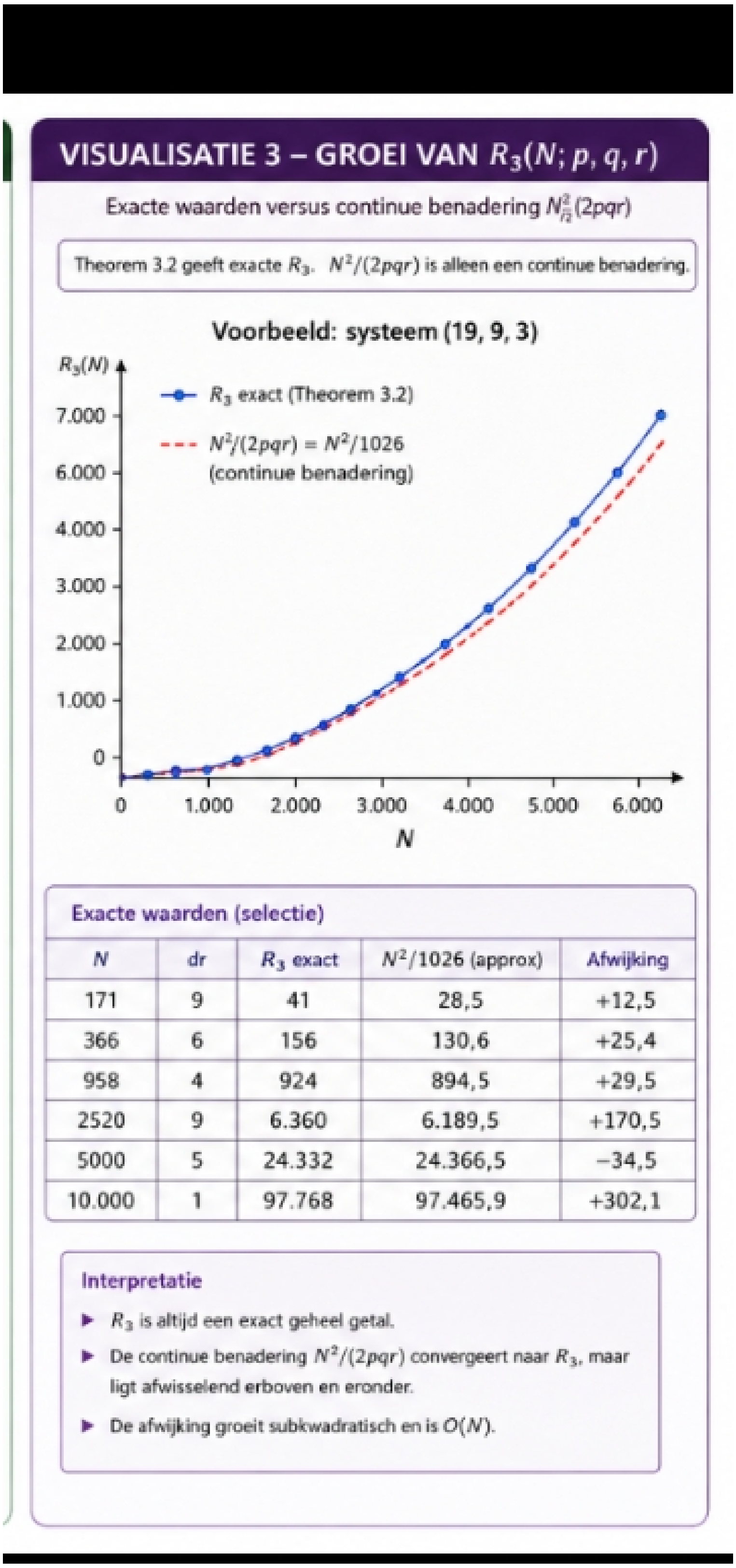

Figure 3 — Exact R■(N) (blue) versus continuous approximation N²/1026 (red dashed). R■ is always an exact integer. The approximation converges to R■ but alternates above and below it.

4.5 Verification across 8 systems

System	period	N	R■ exact	R■ closed	Match
(19,9,3)	3	221	46	46	ok
(10,9,3)	3	140	37	37	ok
(37,9,3)	3	383	68	68	ok
(7,6,3)	2	92	34	34	ok
(7,6,2)	3	92	63	63	ok
(13,4,2)	2	102	66	66	ok
(13,6,3)	2	128	34	34	ok
(7,3,1)	3	71	139	139	ok

§5 The System (19,9,3) and Frobenius Collapse

5.1 Special properties of (19,9,3)

Remark — Frobenius ratio (specific to (19,9,3))

g(19,9)=143, g(19,9,3)=35, ratio=143/35.
This ratio is NOT a general property. Each system has its own:

(10,9,3): 71/17 (37,9,3): 287/71 (7,6,3): 29/11
(7,6,2): 29/5 (13,4,2): 35/11 (13,6,3): 59/23

5.2 Frobenius Collapse

What Frobenius collapse means

The Frobenius number measures how many gaps remain: how large the largest unrepresentable number is.

g(19,9) = 143: with just 19 and 9, there are many gaps.
g(19,9,3) = 35: adding 3 closes most gaps.
g(19,9,3,1) = 0: adding 1 closes all gaps — every N = 1·D.

The Frobenius problem does not reach a limit. It ceases to exist. This is not a termination — it is a collapse.

$$g(19,9) = 143 \to g(19,9,3) = 35 \to g(19,9,3,1) = 0$$

Divisors of r=3: {1, 3}. No non-trivial s with s|3 and s>1 remains.

5.3 A-C cycle for N=958 (dr=4)

j	A	dr(A)	C■■■	A+3C mod 9	= dr(958)?
0	1	1	313	4	yes ✓
1	4	4	294	4	yes ✓
2	7	7	275	4	yes ✓
3	10	1	256	4	yes ✓
4	13	4	237	4	yes ✓
5	16	7	218	4	yes ✓

$A+3C \bmod 9 = dr(958) = 4$ in every row without exception. This is the coupling $A+rC \equiv N \pmod q$ made visible.

§6 Worked Examples

6.1 System (7,6,3), N=92 [period=2]

Symbol	Value	Computation	What it means
a■	2	92 mod 3	A must be ≡2(mod 3): A∈{2,5,8,11,...}
m	13	floor(92/7)	Largest A is 13
M	78	92–77·2=78	Adjusted base
J	3	floor((13–2)/3)	4 valid A-values: {2,5,8,11}
Mq	0	78 mod 6	Starting residue for cycle
period	2	q/r=6/3	Cycle repeats every 2 steps
K	2	floor(4/2)	2 full cycles
t	0	4 mod 2	0 remaining terms
cycle	3	(0–0)%6+(0–3)%6	=0+3=3
R	6	2×3+0	Total remainder correction
R■	34	[4·78–21·3·4/2–6]/6+4	=[312–126–6]/6+4=30+4=34

Verified: r3_exact(92; 7,6,3) = 34. QED

6.2 System (19,9,3), N=958 [period=3]

Symbol	Value	Computation	What it means
a■	1	958 mod 3	A must be ≡1(mod 3)
m	50	floor(958/19)	Largest A is 50
M	939	958–19·1	Adjusted base
J	16	floor((50–1)/3)	17 valid A-values
Mq	3	939 mod 9	Starting residue
period	3	9/3	Cycle length 3
K	5	floor(17/3)	5 full cycles
t	2	17 mod 3	2 remaining terms
cycle	9	(3+0+6)=9	One full cycle sum
R	48	5×9+(3+0)=48	Total correction
R■	924	[17·939–57·16·17/2–48]/9+17	=924

Verified: r3_exact(958; 19,9,3) = 924. QED

§7 Proofs

7.1 Proof of A-C Coupling (Theorem 3.2)

(i) A≡N(mod r): N=pA+rC(mod r): p=1(mod r) by V4, q=0 by V3. So N=A(mod r). QED

(ii) C=(N–A)/r(mod r): from (iii), reduce mod r. QED

(iii) A+rC=N(mod q): N=pA+rC(mod q) since p=1(mod q). QED

(iv) B free: qB=0 mod r and mod q. QED

(v) C■■■=(M–p·r·j)/r: N_rest=M–p·r·j; max C at B=0. QED

(vi) Period=q/r: step=(p·r) mod q=r; period=q/gcd(r,q)=q/r. QED

7.2 Proof of General Formula (Theorem 4.2)

Step 1: Valid A=a■+r·j, j=0..J (J+1 values, by coupling i).

Step 2: Per valid A: count(B,C)=floor((M–p·r·j)/q)+1.

Step 3: R■=Σ floor((M–p·r·j)/q)+(J+1)

=[(J+1)M–p·r·J(J+1)/2–R]/q+(J+1)

where R=Σ■(M–p·r·j) mod q.

Step 4: R has period=q/r by (vi). Write J+1=K·period+t. R=K·cycle+partial. QED

7.3 Proof of Frobenius Collapse

g(19,9,3)=35:

(a) 35 not representable: A∈{0,1}; 3(3B+C)=35 or 16 — not div by 3.

(b) N≥36: bases 36=3×12, 37=19+18, 38=19×2 cover mod-3 classes. Induction: N representable ⇒ N+3 representable (C+1). QED

Frobenius collapse at s=1:

g(19,9,3,1)=0 because every N=1·D=N. Divisors of r=3 are {1,3}. No non-trivial s with s|3 and s>1 exists. The Frobenius problem ceases to exist. QED

§8 Conclusions

Label	Statement	Status
T3.2	A-C coupling: A+rC≡N(mod q) for all valid (p,q,r)	Proved
T4.2	O(1) exact formula R■(N;p,q,r) — replaces Popoviciu recursion	Proved
T7.2	Exact count per A: floor(N_rest/q)+1	Proved
T7.3	Frobenius collapse: g(19,9,3,1)=0, problem ceases to exist	Proved
Vrfy	8 systems verified, 300 values each	Checked
V5	Frobenius ratio 143/35 is specific to (19,9,3), not general	Corrected
Tbl	Deviation in table belongs to $N^2/1026$, not to R■	Clarified

The central result in one sentence:

The condition p≡1(mod q) with r|q forces A+rC≡N(mod q), which collapses the classical O(N/r) recursive Popoviciu sum into an exact O(1) closed-form with period q/r — for every valid (p,q,r), for every N≥pq, without recursion or approximation.

Extensions beyond this paper: (1) exact formula for R■(N;p,q,r,s) for s|r, (2) boundary cases N < pq covered by Q-extension via scaling (Extension I, el Issaoui 2026).

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